

Inflammatory Heart Disease: Diagnosis and Findings of Endocarditis, Myocarditis & Pericarditis

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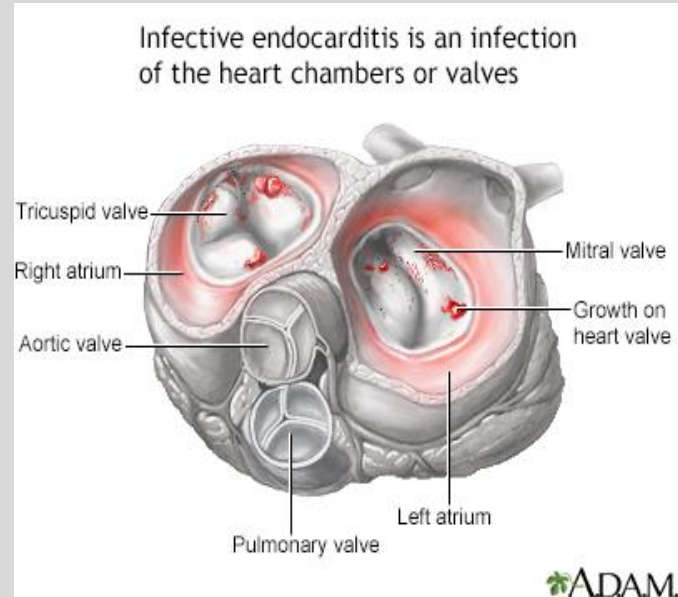
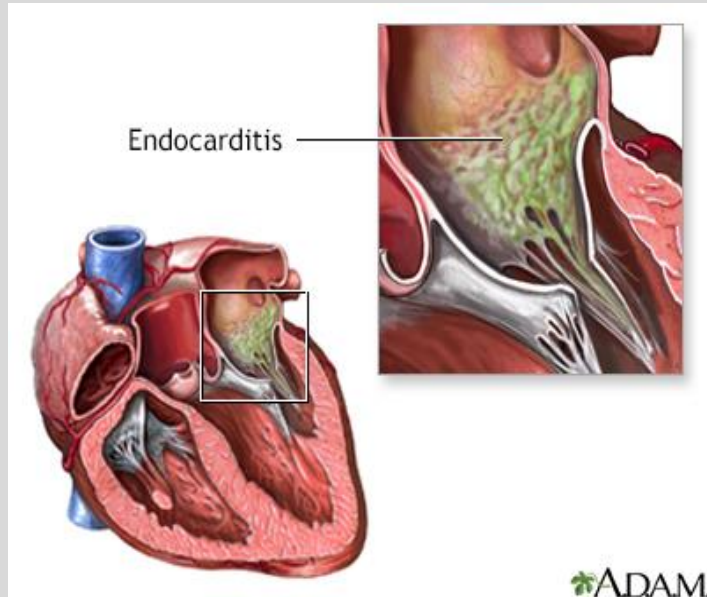
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Session Objectives

1. Describe the etiologies associated with endocarditis, myocarditis, and pericarditis.
2. Explain the clinical presentation of each condition
3. Explain how each condition can be accurately diagnosed.
4. Describe other related clinical forms of inflammatory heart disease, including marantic endocarditis, tamponade, and constrictive pericarditis.

ENDOCARDITIS

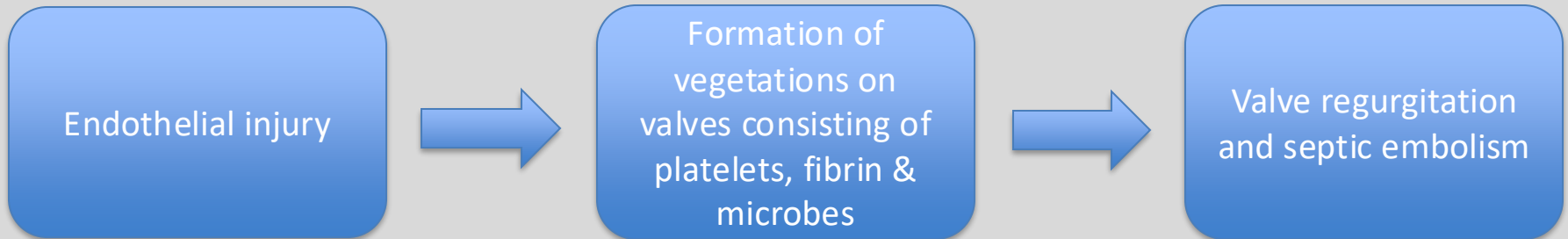
Inflammation of the endocardium that usually involves the cardiac valves.



Risk Factors

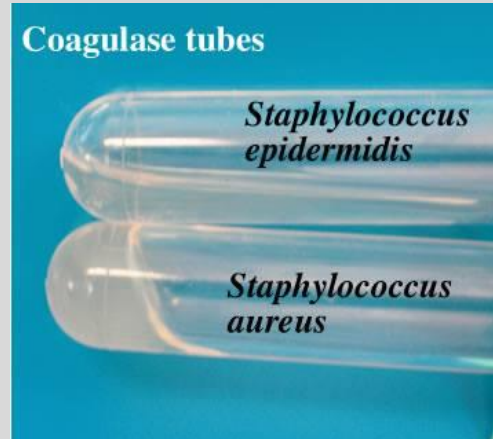
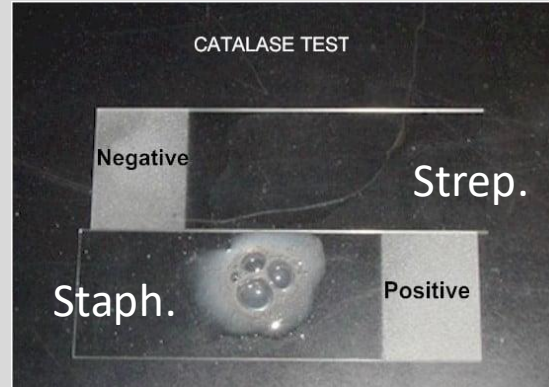
- Age >60
- Injection drug use
- Poor dentition/dental infection
- History of infective endocarditis
- Structural heart disease (valvular, congenital)
- Chronic hemodialysis
- Procedures (intravascular device, cardiac implantable electronic device, cardiac bypass surgery)

Pathogenesis



Staphylococcus

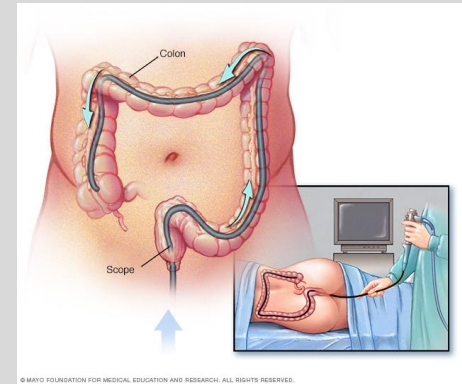
- *Staphylococcus aureus*
 - Gram +, Catalase +, Coagulase +
 - Causes **acute** endocarditis
 - Tends to infect *normal* (native) heart valves
- *Staphylococcus epidermidis*
 - Gram +, Catalase +, Coagulase –
 - Low virulence bacteria found in normal skin flora
 - Associated with prosthetic valves



Streptococcus

- Viridans group *Streptococcus* → *S. sanguinis*, *S. mitis*, *S. mutans*
 - Group of Gram + cocci, Catalase –
 - Found in normal mouth flora; IE may occur after a dental procedure
 - Typically cause subacute (over several weeks) endocarditis that often affect *damaged* heart valves.
- *Streptococcus gallolyticus (bovis)*
 - Gram + cocci found in normal gut bacteria
 - Associated with colon cancer

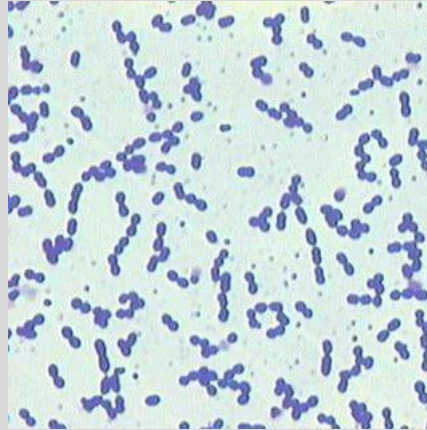
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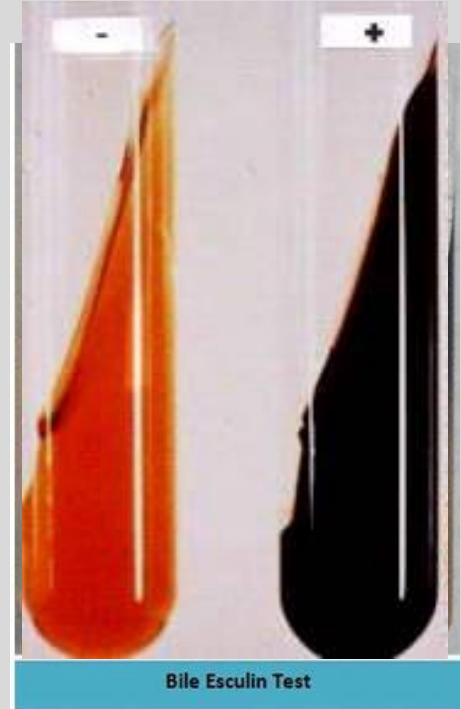
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Enterococcus

- Gram (+) cocci found in normal gut bacteria
- Bile-esculin (+) → agar changes from orange to black
- Enterococcus endocarditis after GI/GU procedures (e.g., abdominal surgery, urinary catheter, TURP, etc.)



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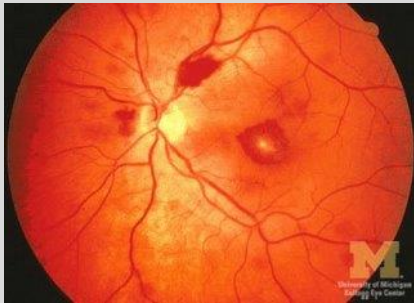
Less common etiologies of IE

- Culture-negative endocarditis → Caused by rare microbes that are difficult to culture:
- Certain Gram-negative bacteria – *HACEK* group; *Coxiella burnetii*, *Legionella*, & *Bartonella*
- Certain Gram-positive bacteria – *nutritionally variant Strep (NVS)*, *Corynebacterium*, *Listeria*
- Fungi – most common ones are *Candida* & *Aspergillus*; other rare ones include *Coccidioides*, *Histoplasma* & *Cryptococcus*

Signs & Symptoms

Fever, murmur, chills, sweating, weight loss, etc.

- Roth spots – Retinal hemorrhage
- Osler nodes – painful bumps on pads of fingers and toes
- Janeway lesions – nontender red macules on palms and soles
- Splinter hemorrhages – reddish-brown lesions under fingernails



Diagnosis of Infective Endocarditis based on Modified Duke criteria:

Patient needs to have one of the following:

- 2 major criteria
- 1 major and 3 minor criteria
- 5 minor criteria

MAJOR:

- Positive blood cultures
- Vegetations on echocardiogram

MINOR:

- Fever ($\geq 38.0^{\circ}\text{C}$ or 100.4°F)
- Risk factors
- Roth spots, Osler nodes, Janeway lesions, splinter hemorrhages

Differential Diagnosis

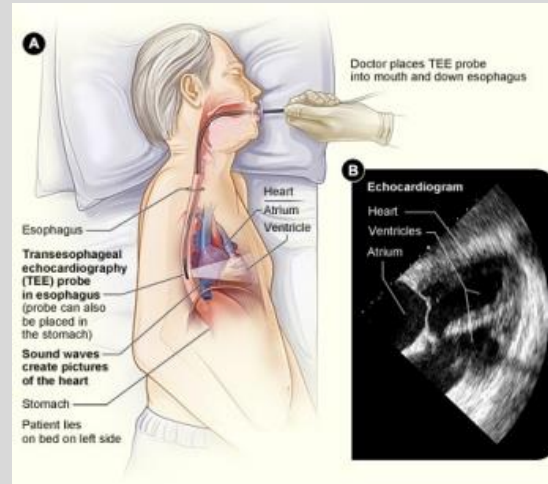
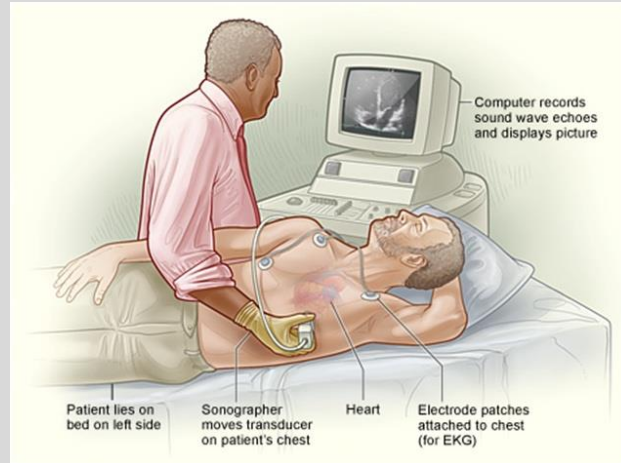
Bacteremia + valvular vegetations = endocarditis

However, in the presence of only one finding, consider other differentials:

- Bacteremia but NO evidence of valvular vegetations → infection from an intravascular catheter, cardiac device, or prosthetic joint, etc.
- Valvular vegetations or arterial emboli but NO bacteremia → culture-negative endocarditis (Libman-Sacks), antiphospholipid syndrome

Workup

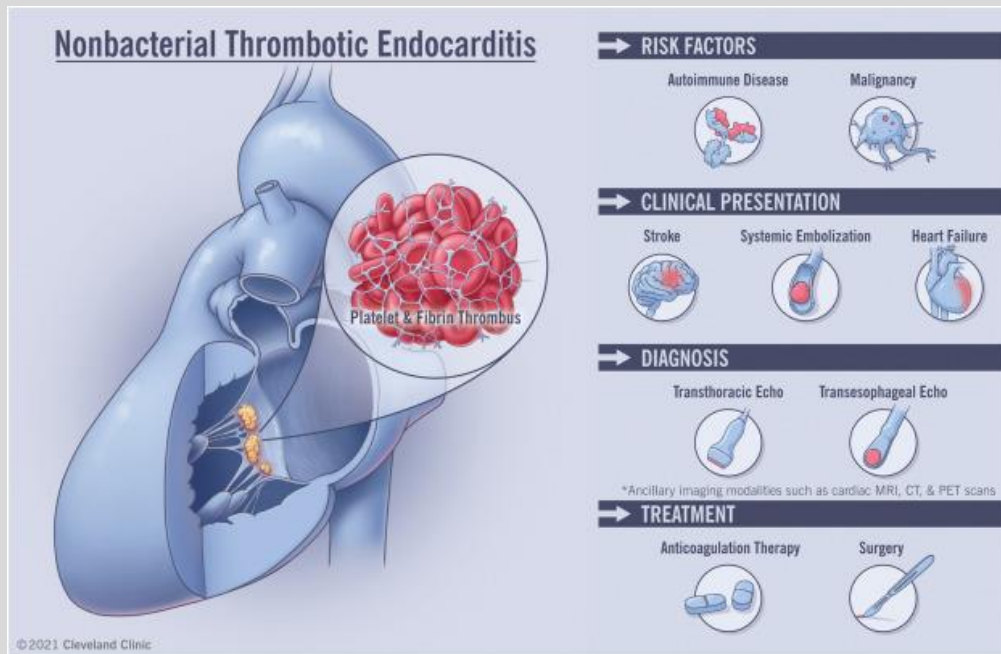
- H&P, ECG, urinalysis (~50% will have proteinuria and microscopic hematuria)
- Lab work: CBC, CMP, PT/INR, PTT, blood cultures (2 sets, preferably 3)
- Diagnostic testing
 - Transthoracic echocardiogram (TTE) or
 - Transesophageal echocardiogram (TEE) – invasive but has higher sensitivity (>90%) and is better detecting cardiac complications



Marantic endocarditis

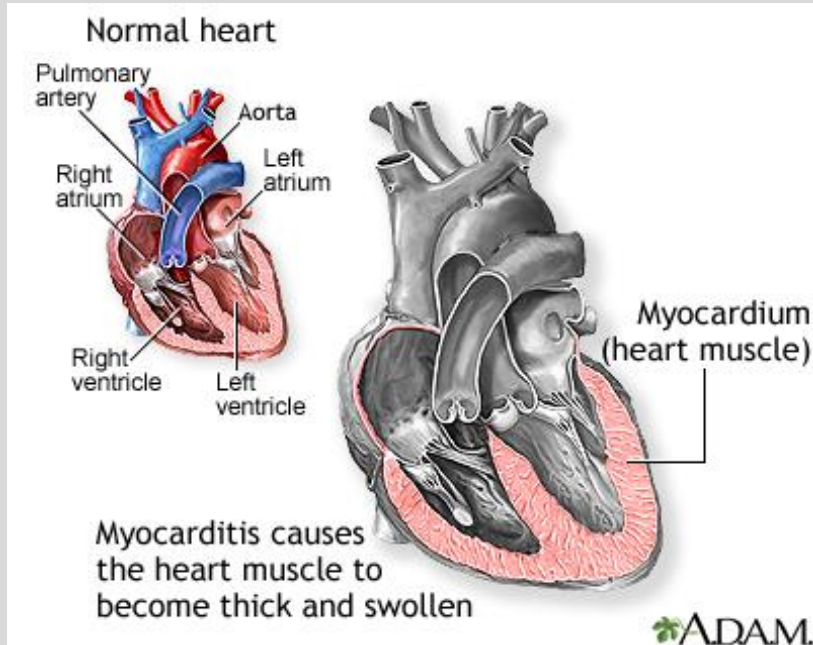
non-bacterial, thrombotic endocarditis

- Characterized by *sterile* vegetations composed of fibrin and platelets
- Seen in hypercoagulable states: malignancy, SLE (Libman-Sacks endocarditis), pregnancy, etc.
- Patients can be asymptomatic. But septic emboli can form.



MYOCARDITIS

Inflammation of the heart muscle with myocyte necrosis.



Etiologies

Inflammation of the myocardium can develop from various causes:

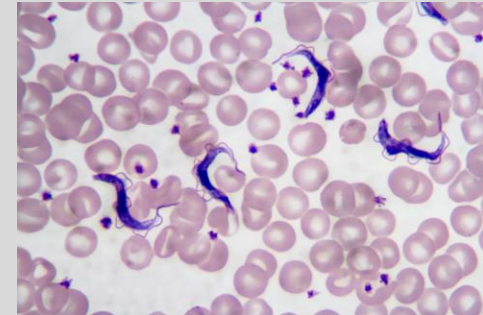
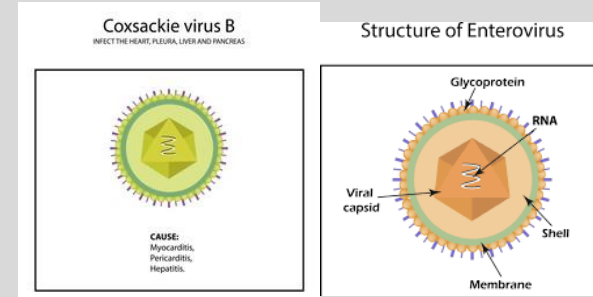
- 50% of cases are idiopathic
- Infection – viral, bacterial, protozoal
- Toxins – ethanol, illicit drug use, chemotherapeutics, environmental toxins
- Immunologic – allergic responses or systemic disorders
- Physical agents – electric shock, radiation, trauma

Microbial Etiologies

The most common causes of myocarditis in *North America and Europe* = **viral etiology** – parvovirus B19, enterovirus, adenoviruses, SARS-CoV-2, HHV6, Influenza, etc.

The most common cause of myocarditis *worldwide* is the protozoan ***Trypanosoma cruzi*** → Chagas disease

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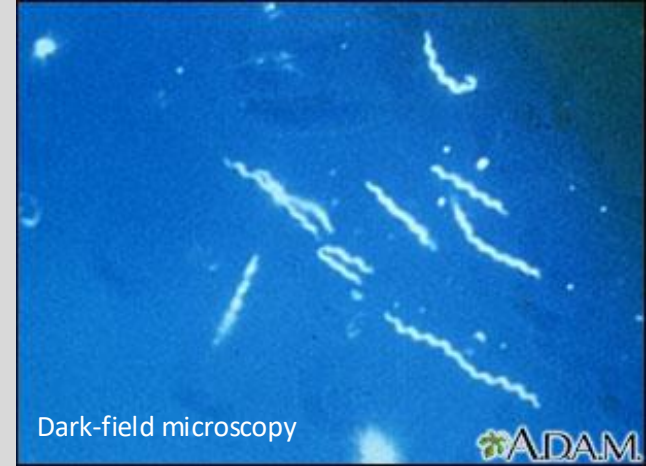


Lyme Carditis

Caused by the spirochete *Borrelia burgdorferi*

Stage 1: early localized infection

Erythema migrans from tick bite



Stage 2: early disseminated infection
(days to months later)

Transient colonization in bloodstream
may cause secondary skin lesions, AV
block, myocarditis

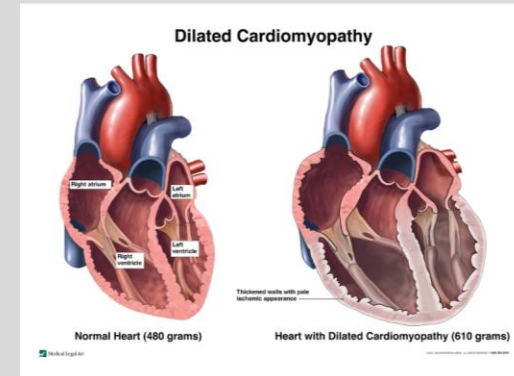


Stage 3: late stage infection (months to years later)

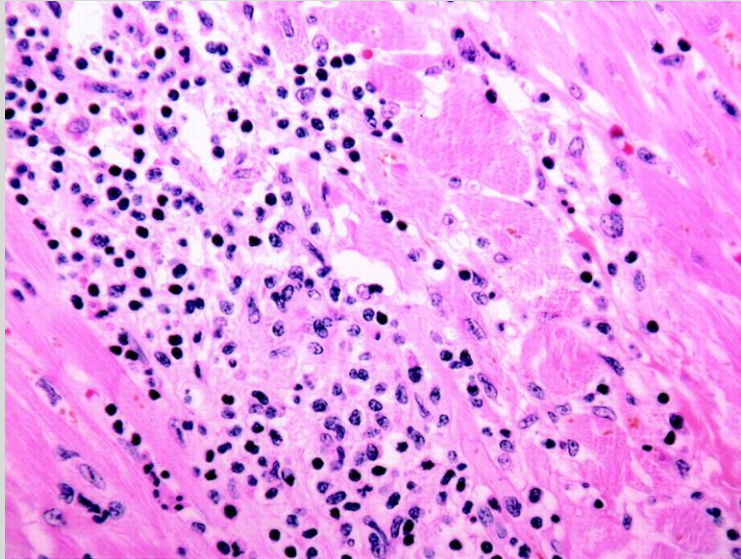
Chronic infection in multiple tissue sites can cause
Lyme arthritis, neurological issues

Clinical Presentation

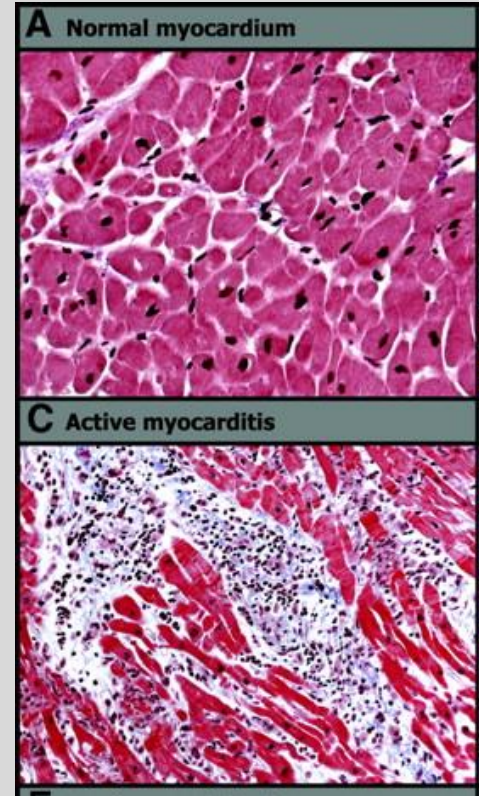
- Patient *may be asymptomatic*, or symptoms may be similar to the flu:
 - Fever, chest pain and other constitutional symptoms (e.g., headache, fatigue, muscle aches)
- Myocarditis is rarely fulminant
- Complication: Chronic myocarditis can cause persistent ventricular dysfunction, which can lead to **heart failure (dilated cardiomyopathy)**



Dallas criteria defines active myocarditis as “an inflammatory infiltrate of the myocardium with necrosis and/or degeneration of adjacent myocytes not typical of the ischemic damage associated with coronary artery disease.” The infiltrates are usually mononuclear, but may be neutrophilic, or occasionally, eosinophilic

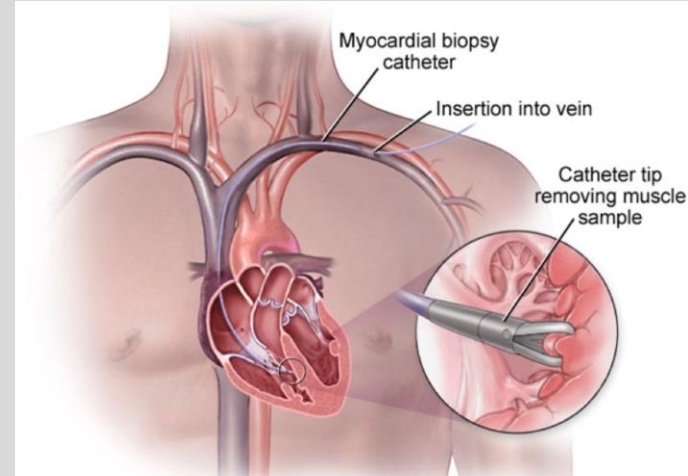
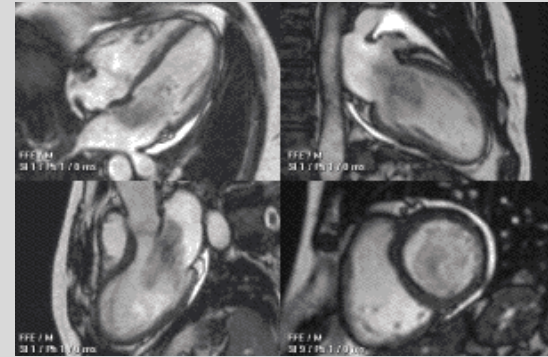


Lymphocytic (viral) myocarditis. Endomyocardial biopsy with hematoxylin and eosin; magnification $\times 400$. Text and image courtesy of James R. Stone, MD, PhD.



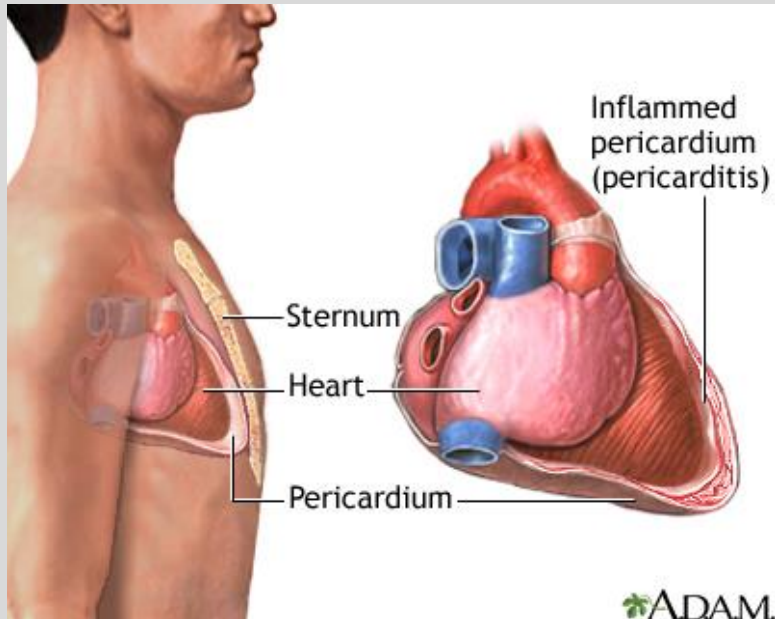
Workup

- Initial evaluation: ECG, troponins, CBC, inflammatory markers (CRP, ESR), echocardiogram
- Gold standard noninvasive test = **cardiac magnetic resonance imaging (CMR)**
- Gold standard diagnostic test = **Endomyocardial biopsy (EMB)**
 - Indicated in select cases, such as unexplained fulminant heart failure, life-threatening arrhythmias, etc.
 - You cannot make a definitive diagnosis of myocarditis without EMB that meets Dallas criteria
- Treatments and interventions should be individualized based on clinical context and comorbidities

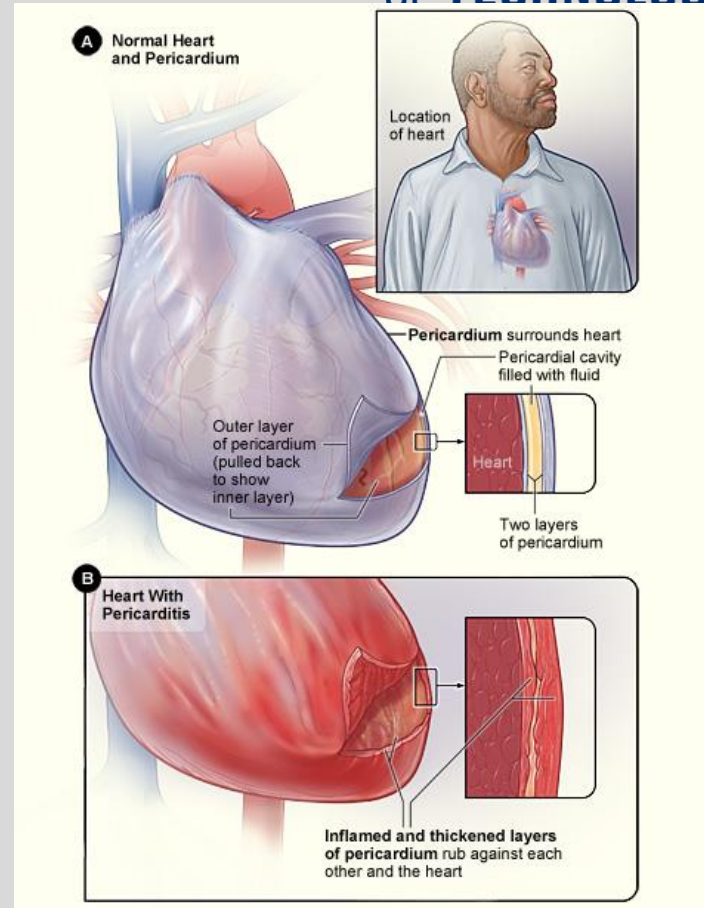


PERICARDITIS

An inflammation of the pericardium.



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Etiologies

- Etiology of acute pericarditis is **usually idiopathic**
 - Enterovirus, adenoviruses, SARS-CoV-2, HIV, Influenza, parvovirus B19, etc.
- Other causes:
 - Bacteria: *Streptococcus pneumoniae*, *Staphylococcus aureus*
 - Tuberculosis: Subacute, chronic. *Mycobacterium tuberculosis* is the most frequent cause worldwide
 - Fungi

Clinical Presentation & Diagnosis

- Symptoms:
 - Chest pain – sharp, worsens with inhalation and lying supine, better when sitting up/leaning forward
 - Fever
- **Pericardial friction rub** – a scratchy sound heard in systole and diastole
- ECG changes - ST elevation, PR depression
- Pericarditis can involve **pericardial effusion** – excess fluid (serous, blood, pus, or lymphatic fluid) in the pericardial space



Cardiomegaly due to a massive pericardial effusion. At least 200 mL of fluid causes the silhouette to enlarge.

Cardiac Tamponade

Tamponade = high pericardial pressure compressing the heart

- Fluid around the heart = pericardial effusion → When the effusion compresses the heart → cardiac tamponade
- Decreased cardiac output
- Clinical presentation: Beck's triad = hypotension, jugular venous distention (JVD), muffled heart sounds

Constrictive Pericarditis

Chronic inflammation and fibrous constriction of the pericardium

- Pericardial fibrosis and calcification restricts space for expansion → decreased ventricular filling
- Clinical presentation: dyspnea, Kussmaul sign (increased jugular venous pressure), pericardial knock (high-pitched sound in early diastole)
- Most common cause resource-limited countries = **Tuberculosis**

Summary

- **Endocarditis:** Infection of endocardium/valves; often *S. aureus*, *Strep viridans*, enterococci. Risk factors: IV drug use, prosthetic valves, structural heart disease. Related: Marantic endocarditis
 - Clinical: Fever, new murmur, embolic/immune lesions (Roth spots, Osler nodes, Janeway lesions)
 - Dx: Duke's criteria, blood cultures, TTE/TEE
- **Myocarditis:** Inflammation of myocardium; often viral (can be bacterial or protozoal)
 - Clinical: Variable; can mimic MI; can lead to cardiomyopathy
 - Dx: Imaging ± biopsy (Dallas criteria)
- **Pericarditis:** Inflammation of pericardium; often viral (can be bacterial or fungal)
 - Clinical: Sharp chest pain, better leaning forward; friction rub; ECG change
 - Dx: Clinical + ECG, imaging to assess effusion
 - Complication: tamponade, chronic constriction

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